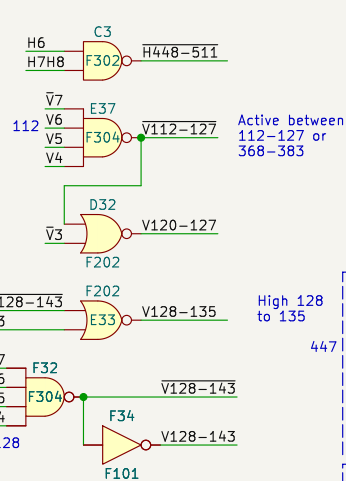
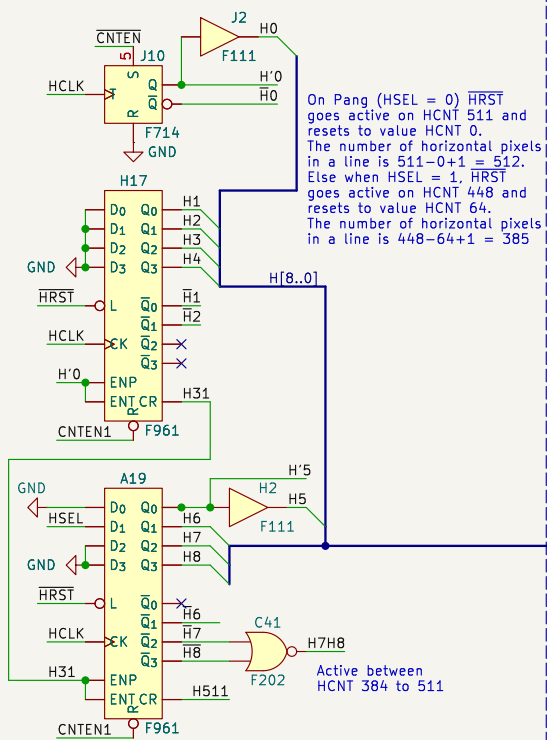
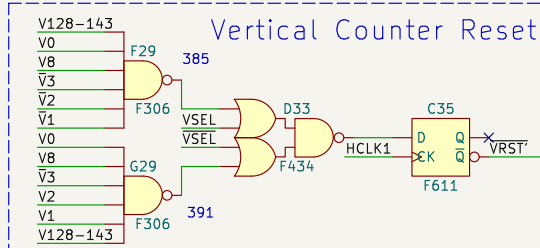


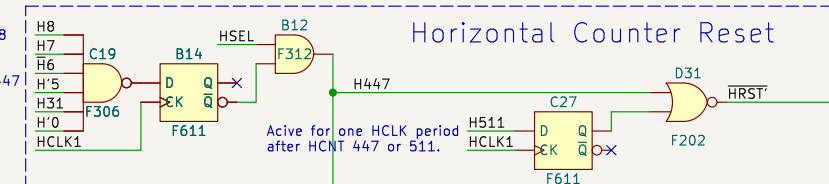
## Horizontal Counter



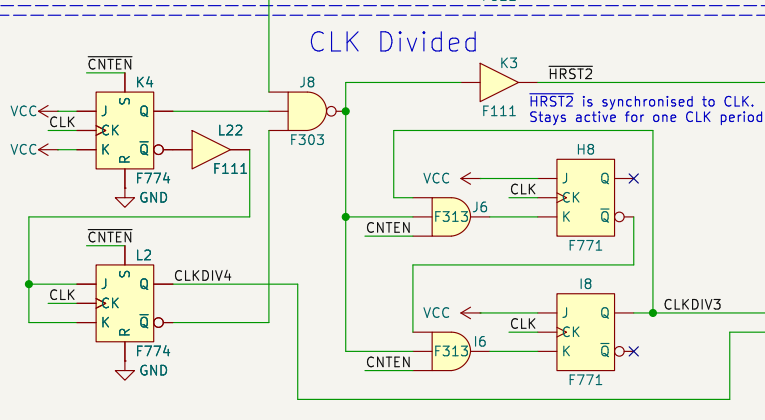
## Vertical Counter Reset



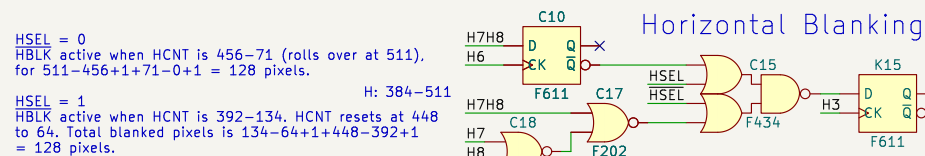
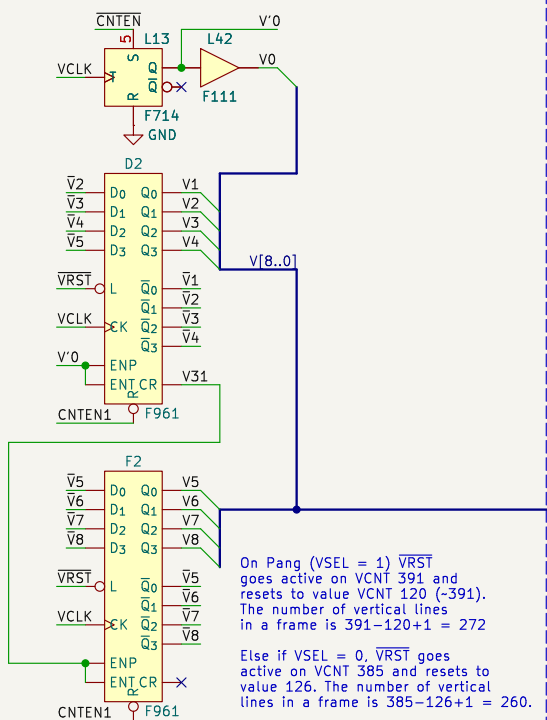
## Horizontal Counter Reset



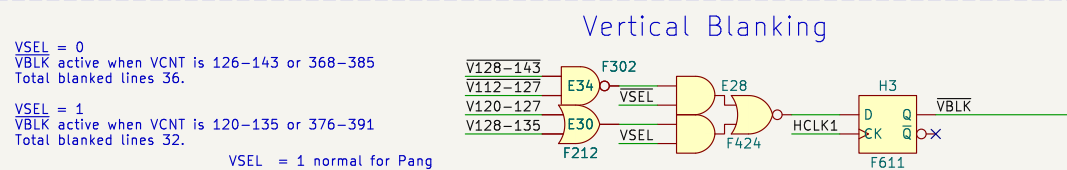
## CLK Divided



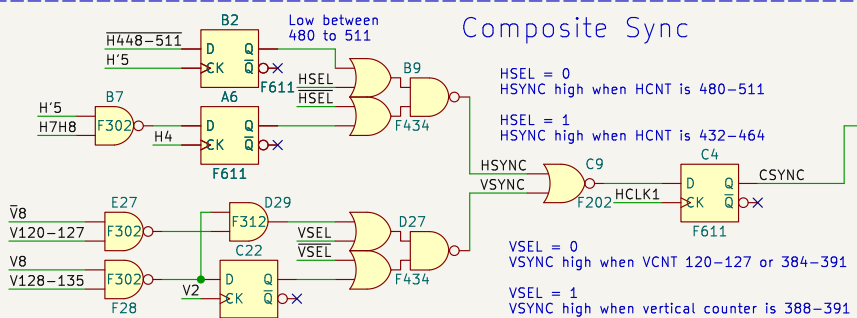
## Vertical Counter



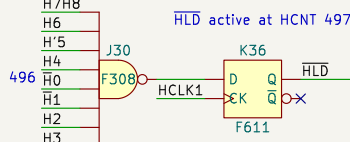
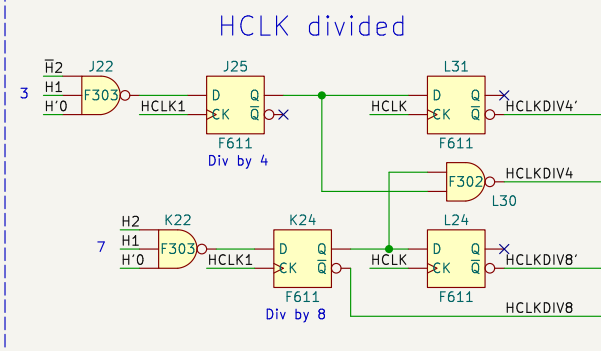
## Vertical Blanking



## Composite Sync



## HCLK divided



## Timing

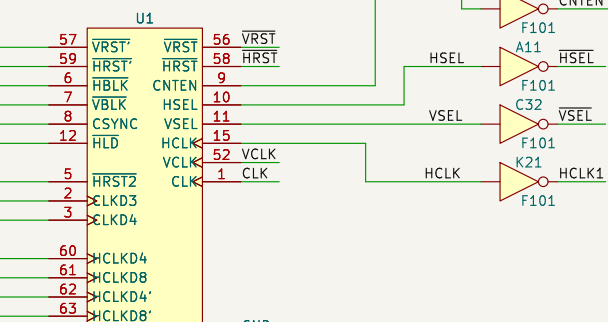
File: timing.kicad\_sch

The Schematics have been analyzed from a die picture by InfoSecDJ.  
The chip is identified as:

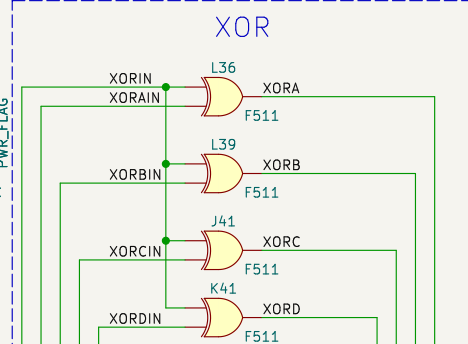
NEC  
65006-103

It is a uPD65000 (CMOS-2) series, 3-micron gate array.  
- 528 Basic Cells  
- 12 Columns, 44 Rows

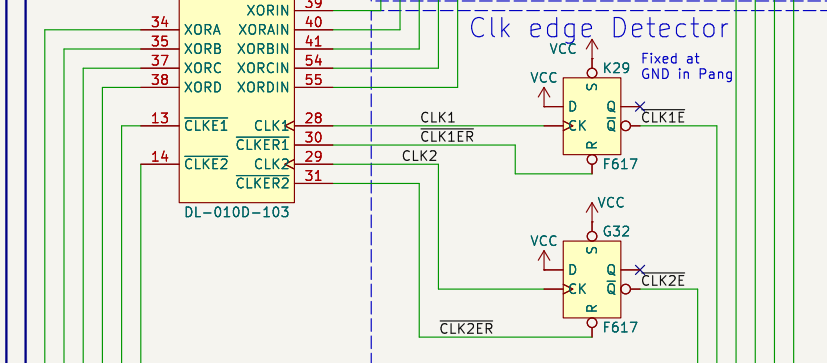
CNTEN Connected to VCC on pang



## XOR



## Clk edge Detector



Ulf Skutnabba, twitter: @skutis77

Sheet: /  
File: dl-010d-103.kicad\_sch

**Title: Capcom DL-010D-103**

Size: A3 Date: 2025-09-18  
KiCad E.D.A. 8.0.9

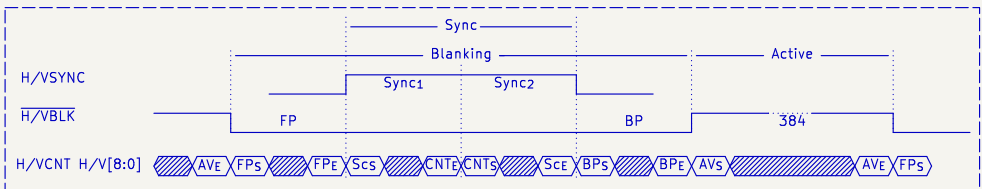
Rev:  
Id: 1/2

|                                    |     |     |
|------------------------------------|-----|-----|
| HSEL                               | 1   | 0   |
| FP <sub>S</sub> ,HBLK <sub>S</sub> | 392 | 456 |
| FP <sub>E</sub>                    | 431 | 479 |
| SC <sub>S</sub>                    | 432 | 480 |
| CNT <sub>E</sub>                   | 447 | 511 |
| CNT <sub>S</sub>                   | 64  | 0   |
| SC <sub>E</sub>                    | 79  | 31  |
| BP <sub>S</sub>                    | 80  | 32  |
| BP <sub>E</sub> ,HBLK <sub>E</sub> | 135 | 71  |
| AV <sub>S</sub>                    | 136 | 72  |
| AV <sub>E</sub>                    | 391 | 455 |

### Horizontal/Vertical Counter Abbreviations

|             |                                        |
|-------------|----------------------------------------|
| AVS, AVE:   | Active Video Start/End                 |
| FPS, FPE:   | Front Porch Start/End                  |
| BPS, BPE:   | Back Porch Start/End                   |
| CNTS, CNTE: | Horizontal/Vertical Counter Start/Stop |
| SCS, SCE:   | Horizontal/Vertical Sync Start/Stop    |

Active: Active video signal,  $AV = AV_E - AV_S + 1$   
 Blanking: Blank video signal  
 Sync: Synchronizing video signal  
 FP: Front Porch,  $FP = FP_E - FP_S + 1$   
 BP: Back Porch  
 $Sync = Sync_1 + Sync_2$



|                                              |     |     |
|----------------------------------------------|-----|-----|
| VSEL                                         | 0   | 1   |
| FP <sub>S</sub> , $\overline{\text{VBLK}}_S$ | 382 | 376 |
| FP <sub>E</sub>                              | 383 | 387 |
| SC <sub>S</sub>                              | 384 | 388 |
| CNT <sub>E</sub>                             | 385 | 391 |
| CNT <sub>S</sub>                             | 126 | 120 |
| SC <sub>E</sub>                              | 127 | 123 |
| BP <sub>S</sub>                              | 128 | 124 |
| BP <sub>E</sub> , $\overline{\text{VBLK}}_E$ | 143 | 135 |
| AV <sub>S</sub>                              | 144 | 136 |
| AV <sub>E</sub>                              | 381 | 375 |

HSEL = 0, VSEL = 1 is used on all games on the Capcom/Mitchell platform.

Total pixels: 512  
Active pixels: 384  
Blanked pixels: 128  
Pixel clock: 8Mhz  
Hsync frequency: 15.625kHz  
Vertical lines: 272  
Blanked lines: 32  
Vsync frequency: 57.44485Hz